RDF Store Vendor Evaluation & Scorecard

Prepared for PEO Architecture Team review · Nov 3, 2026 · @jeffrey

1. Summary

Cambridge Semantics Anzo and Stardog are the two top contenders and are recommended to advance to a scored comparison against the capability list in Section 4. Six vendors were reviewed for the semantic modeling and RDF store layer behind the Semantic Knowledge Layer (SKL). PoolParty is not recommended, TopBraid EDG would need significant build-out, and Ab Initio and Alation were assessed as companions to Protégé rather than as full RDF stores.

Vendor	Meetings	Status	Bottom line
Cambridge Semantics Anzo	3/20, 3/21	**Top contender: advance**	Automated mapping, top-down and bottom-up modeling, and professional services proven on large, complex semantic layers.
Stardog	3/14, 3/18, 3/20	**Top contender: advance**	Knowledge graphs without a heavy transformation lift, solid services, and an RDF store option if MarkLogic is cut.
TopBraid EDG	3/14, 3/15	Conditional	Governance is built in, but it needs extra build-out on our side (no accelerators) and services lack experience at our scale.
Ab Initio	Not recorded	Possible, heavy workaround	Could work alongside Protégé, but launch would be difficult without a dedicated support team.
Alation	Not recorded	Review incomplete	Strengths and bottom line were left blank in the source review.

Vendor	Meetings	Status	Bottom line
PoolParty (Semantic Web)	3/21	Not recommended	Application-building focus rooted in the relational world; feels like a repeat of Semaphore.

2. Vendor reviews

2.1 Cambridge Semantics Anzo: top contender

Met 3/20 and 3/21.

Strengths

- **Modeling:** top-down and bottom-up modeling. Models are auto-generated at import from observed metadata (schema, etc.) and can ingest external ontologies, including UML in XML format, which must be mapped to the Anzo-generated models.
- **Ingestion and mapping:** automated mapping; catalog and models generated at data import; metadata can be read from the data catalog.
- **Built-in functionality:** dashboards, APIs created within the interface, data and table views, a modeling area, and geographical visualizations.
- **Data handling:** data is not stored in Anzo. It is brought into memory only when a graph mart (the instructions used to build a knowledge graph) is created or run.
- **Services:** honest about capabilities, with knowledgeable implementation staff. The senior professional services team has delivered large, robust semantic layers for large, complex companies.

Concerns

- Unclear how to integrate business users into the process.
- Validation and rules need to be applied judiciously.

Bottom line: top contender; move forward to next steps.

2.2 Stardog: top contender

Met 3/14, 3/18 and 3/20.

Strengths

- **Knowledge graphs in practice:** real application without a heavy lift to transform data. Inference is computed at query time.
- **Fit with our situation:** its RDF store could be used if MarkLogic is cut, and implementation staff understand our difficulties.
- **Catalog and tuning:** catalog integrations are available (Stardog can act as a catalog of catalogs). The Query Profiler automatically generates SQL to help with tuning and optimization, and a caching mechanism is available for querying.
- **Services:** solid, with the experience needed for our scale and complexity.

Concerns

- Cloud-only implementation, which may raise security and risk management concerns.
- Integration and performance may become limiting as we scale.
- Visualization is limited for managing taxonomies and working with hierarchies.

Bottom line: top contender; move forward to next steps.

2.3 TopBraid EDG: conditional

Met 3/14 and 3/15.

Strengths

- Governance is built into the structure, with change history available.
- IRIs for instances, with global settings available in EDG.
- Highly flexible in how different data is referenced.

Concerns

- The implementation felt simplistic. TopBraid works with the semantic models only, not the instances.
- No visual modeling; the underlying code has to be manipulated.
- Graph Explorer offers simple, customized visualizations similar to Protégé, with no filtering.

Bottom line: it could work but would need additional build-out on our part (no accelerators), and its professional services lack experience at the scale and complexity we need.

2.4 Ab Initio: possible, with heavy workaround

Evaluated for use in conjunction with Protégé.

Strengths

- Glossaries and taxonomies look relatively simple to manage.
- Could accelerate and scale semantically informed microservices, though the integration architecture is on hold for Corporate Data & Ecosystem.
- Could replace the Metrics Catalog and Metrics Engine.
- Bi-directional integration with the SKL is possible (use cases exist), though only for a subset of the SKL.

Concerns

- **Evaluation limits:** the reviewer had no documentation showing what the tool would look like, and creating anything in the proof of concept required another person's approval because of the reviewer's permission level.
- **Governance:** all created content is shared across the platform, so any user with higher permissions could change attributes or content. This open semantic hub approach needs further discussion with the vendor.
- **Standards:** its idea of an ontology may differ from ours, and it is not based on W3C standards.
- **Scope:** no support for mapping to RDF or content types beyond the metrics and context drawn from sources and structures used to deliver analytics. It focuses on defining and maintaining hierarchies for analytics and on inferring relationships (Semantic Discovery) for terms culled from data structures.

Bottom line: could possibly work, but much work-around is needed and launch would be difficult without a dedicated support team.

2.5 Alation: review incomplete

Evaluated for use in conjunction with Protégé.

Strengths: not recorded in the source review.

Concerns

- dbt's semantic layer focuses on mining context about the metrics lifecycle and on improving reuse and search (finding the right metric).
- Bi-directional integration with the SKL is possible, but would support only a subset of the SKL.
- May be able to replace the Metrics Catalog.

Bottom line: not recorded in the source review.

2.6 PoolParty (Semantic Web): not recommended

Met 3/21.

Strengths

- Results can be shared externally in a practical way, which suits content applications.

Concerns

- Tooling and services focus on building applications on a semantic layer curated from the relational (RDB) world.
- Bottom-up structure, firmly rooted in the RDB approach of building apps on top of existing structures and tools.

Bottom line: not a fit; it feels like Semaphore redux.

3. Requirements for the short list

The requirements list is still to be refined. The vendor questions below should be answered for each finalist, followed by the minimum and nice-to-have capabilities.

Vendor experience to confirm

- Experience implementing formal knowledge engineering in companies.
- Examples of the lifecycle from taxonomies to ontologies to knowledge graphs to context mapping to content.
- Experience mapping enterprise semantics to isolated islands of semantics across data repositories, processes and systems.
- Guidance on ingesting metamodels and/or catalogs to support reverse engineering and republishing from the enterprise semantic layer.

Minimum requirements

- Compliance with ISO/NISO standards.
- Models W3C ontologies (OWL) and W3C vocabularies (SKOS).
- Triples can be exported and used by other tools without large data manipulation effort.
- Simple integrations to import content from other resources and to export content to other resources, such as third-party RDF stores.
- Creates and manages concept-based schemes.
- Creates and manages terminology-based vocabularies such as thesauri and taxonomies.

- Creates and manages inference-bearing class-subclass hierarchies.
- Lets different user communities curate alternative sets of preferred terminology for a common conceptual base.
- Unique identifiers (GUIDs and URIs/IRIs) for all concepts.
- Shows a concept's relationships as a flat list, a hierarchical list, or a graph visualization.
- Simple and advanced search modes.
- Manages multiple project teams and assigns users and taxonomies.
- Granular, per-project control of user permissions.
- Shares selected taxonomies with non-editorial stakeholders.
- Batch editing, archiving and versioning tools.
- Multiple data formats for viewing, importing and exporting vocabularies, including multiple RDF serializations and CSV, TAB and Excel files.
- User account management and role-based permissions.

Would like to have

- Tool-agnostic URI/IRI generator.
- Automatic prevention of term duplicates (disambiguation).
- Automatic inverse relationships.
- Automatic prevention of circular references.
- Support for poly-hierarchical structures.
- Import of candidate term lists from spreadsheets and text files.
- Creating new concepts without leaving the relationship-editing workflow.
- Browsing alphabetically, hierarchically and by visualization graphs.
- Custom workflow and governance states to control the flow of work.
- Management reports that can be shared and reused by other users.
- Tracking of all editorial activity by activity type, date-time and user ID.

4. Scorecard: Anzo vs Stardog

The source note suggests scoring only the top two vendors against this list, so the scorecard covers Anzo and Stardog. **No scores have been assigned yet.** The scale below is a proposal for the team to confirm before scoring starts.

Score	Meaning
0	Not supported
1	Partial, or requires custom build-out
2	Supported out of the box
3	Strong: native, and proven at our scale and complexity
N/A	Not applicable; exclude from the average

Team decision needed: whether categories carry equal weight or are weighted by importance. Evidence from vendor demos, documentation or proofs of concept goes in the Notes column.

A. Functional

A1. Vocabulary management

Capability	Anzo	Stardog	Notes / evidence
Text analysis and text extraction			
Semantic associations			
Attributes and properties on nodes and edges			
Categorizing edges			
Mining new terms from imported RDF and OWL structures			
Weights on edges (especially for rules)			
Reasoning and inference rules (RIF), leveraging terms from SPIN or SHACL			
Instantiating math symbols (calculations, rules, etc.)			
Constructing consistent URIs and IRIs			

Capability	Anzo	Stardog	Notes / evidence
Universal identifier for all data and things, independent of labels and properties			
Category average			

A2. Taxonomy management

Capability	Anzo	Stardog	Notes / evidence
Create SKOS standard vocabularies			
Auto tagging and semantic extensions			
Organize entities into hierarchies			
Build associated taxonomies			
Manage taxonomies			
Direct and indirect inheritance			
Change management			
Version management			
Workflow and control			
Impact analysis			
Publish or promote taxonomies			
Warning messages when moving or pruning nodes			
Category average			

A3. Ontology management

Capability	Anzo	Stardog	Notes / evidence
Define patterns in ontologies			
Evolve taxonomies into a knowledge graph			
NLP-supported search			

Capability	Anzo	Stardog	Notes / evidence
Deprecate an ontology without deleting it			
Declarative query language (SPARQL, GraphQL, Cypher, etc.)			
Augmented search			
Navigate structured and unstructured data			
Harmonize and query content from multiple sources			
Evolutionary development and engineering			
Match and align ontologies			
Match and align schemas			
Match and align semantics			
SHACL constraints and rules construction			
Category average			

A4. Relationship management

Capability	Anzo	Stardog	Notes / evidence
Meta properties (attributes and properties on relationships)			
Event relations (n-ary support)			
Extract information from entities and relations using inferred meanings			
Extract information from entities and relations using disambiguation			
Category average			

A5. Data lifecycle management

Capability	Anzo	Stardog	Notes / evidence
Manage NULLs and duplicates in data			
Transform data by matching, fusing, merging, etc.			
Support and enforce the knowledge capture methodology (discovery, vocabulary, taxonomy, ontology)			
Support and enforce the implementation methodology (validation, mapping, query, measurement)			
Capture inputs, processing and outputs across the lifecycle			
Impact analysis and simulation			
Incremental and iterative development			
Category average			

A6. Visualization management

Capability	Anzo	Stardog	Notes / evidence
Visualize data in tree form			
Optimize the number of nodes			
Optimize weights on nodes			
Filtering			
Manage and run NLP queries			
Optimization and pathing			
Exploration			
Category average			

A7. Model capabilities and digital twins

Capability	Anzo	Stardog	Notes / evidence
Multimodal graph database (RDF + LPG)			
Transaction model (ACID)			
Digital twin-specific use cases			
Category average			

B. Non-functional

B1. Performance and extensions

Capability	Anzo	Stardog	Notes / evidence
Scale performance linearly and horizontally			
Centralized and distributed graph databases			
On-prem and off-prem (multiple clouds)			
Location-based intelligence			
AI extension			
Data integration extension			
Analytics and aggregation extension			
Modeling and analysis extension			
Recommendation engine extension			
Semantic data integration extension			
Social network analysis extension			
Category average			

B2. Scalability

Capability	Anzo	Stardog	Notes / evidence
Scale without limit to store and manage large data volumes			
Scale horizontally and vertically			
Auto scaling in AWS/GCP			
Kubernetes in AWS/GCP			
Concurrent users (about 50) running searches			
Category average			

B3. Integration (import/export)

Capability	Anzo	Stardog	Notes / evidence
API support for bi-directional formats: RDF, RDFS, OWL, XSD, SKOS, JSON			
Security support using LDAP and RBAC			
Bi-directional RDB and RDF integration via RDF2RDB mapping			
Bi-directional RDB and RDF integration via R2RML for RDF			
Bi-directional RDB and RDF integration via SQL queries			
Bi-directional RDB and RDF integration via SPARQL queries			
Bi-directional RDB and RDF integration via virtual mapping (SQL2RDF)			
D2R optimization (for querying the RDB)			
Category average			

B4. Security

Capability	Anzo	Stardog	Notes / evidence
Integrate with IdP providers for enterprise SSO and authentication			
Role-based access control			
Monitor, track and audit access control of the system			
Add additional owners and/or editing rights to assets			
Grant different access levels to assets for granular user permissions			
Category average			

B5. Compliance

Capability	Anzo	Stardog	Notes / evidence
GDPR and CCPA data privacy requirements (accountability, data minimization, lawful processing)			
Standards such as ISO/IEC 27001/27002, GSAM and FISMA			
Audit trail of data access and modifications by user ID and date-time			
Category average			

B6. Availability, backup and recovery

Capability	Anzo	Stardog	Notes / evidence
Active-active server configurations during cluster failover and disaster recovery			
99.99% uptime SLA for operational support			

Capability	Anzo	Stardog	Notes / evidence
Fail-safe measure if the database host fails			
Store and back up assets for DR or cluster failover			
Business continuity using backup/restore process			
Business continuity using transaction recovery process			
Centralized backup/recovery catalog across multiple databases			
Failover from primary to standby on unplanned primary DB failure			
Data replication			
Add or remove a standby/DR database without impacting the primary			
Category average			

B7. Support

Capability	Anzo	Stardog	Notes / evidence
24/7 support			
Category average			

5. Governance recommendation and next steps

The Governance team recommends a two-track approach: a semantic modeling tool from the top, and open standards for RDF stores from the bottom.

- **Top-down:** adopt the semantic modeling tool in partnership with business stakeholders, not technical staff alone. The Knowledge Engineering (KE) team drives and facilitates the SKL.
- **Bottom-up:** require open standards for RDF stores plus IRI generation and preservation, so one or more RDF stores can be used over time, perhaps starting with the initial footprint from the top-down vendor.

Proposed SKL architecture

The source describes this as a three-tier architecture, followed by two supporting areas.

1. **Semantic Knowledge Layer (SKL):** the domain fabric.
2. **Unified Data Model:** the RDF fabric.
3. **Event-driven architecture:** loading into RDF stores, orchestration, and discrete services (data, analytics, AI, LLM).

Learning

- Recursive training for domain LLMs using domain ontologies.
- RAG training for properties and relationships, at varying levels of granularity within the SKL.
- Controlled inferencing for exploration and impact (scenario extension, cached only).

Loading

- Sources of truth (SOT): metamodel, catalog, Ab Initio, Collibra, Engram, etc.
- Systems of record (SORs): data, documents, metrics, analytics, AI and LLMs.

Open items before the decision

- Agree the scoring scale and any category weighting (Section 4), then score Anzo and Stardog.
- Anzo: clarify how business users will be brought into the modeling process.
- Stardog: assess the cloud-only deployment against security and risk management requirements, and test integration and performance at scale.
- Ab Initio: hold the further vendor discussion on governance and ontology standards; the integration architecture is currently on hold for Corporate Data & Ecosystem.
- Alation: complete the review, since strengths and bottom line are blank.
- Confirm the MarkLogic decision, since it affects the value of Stardog's RDF store.
- Refine the requirements list (Section 3) and confirm which items are hard minimums.

External reviews

- Stardog: awards, Gartner Peer Insights, G2 reviews
- Cambridge Semantics Anzo: company news, Gartner Peer Insights, G2 reviews